INCH-POUND
A-A-59422A
April 10, 2025
SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

PUNCHING, SHEARING, NOTCHING AND COPING MACHINE, METAL, MECHANICAL DRIVE, SINGLE-OPERATOR IRONWORKER

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers mechanically driven metalworking machines used for performing shearing, punching, bending, notching, and coping operations on bar stock, a wide variety of structural shapes, and plates material, primarily metal. The term "ironworker" as used in the remainder of this document refers to the system in its entirety including hydraulic and electrical systems, working surfaces, machine capacities, and any defined peripheral equipment. Ironworkers identified herein shall be manufactured within the United States of America and shipped as a complete unit.

2. CLASSIFICATION. The ironworker shall be classified and identified by type in accordance with the defining characteristics specified in Table I. The type of ironworker to be supplied shall be specified in the acquisition order (see Table I).

TABLE I. Product classifications.

TYPE	Capacity (tons)	Throat depth (inches)
I	50	5
II	90	10
III	110	12

3. SALIENT CHARACTERISTICS.

3.1 Design. The ironworker shall accomplish not less than the five operations of shearing, punching, bending, notching, and coping. All stations shall be powered by the same drive. The frame and base of the ironworker shall be an integral unit designed to rest on a flat surface. The ironworker shall be manufactured in accordance with American National Standards Institute (ANSI) B11.5, Ironworkers, Construction, Care, and Use, ANSI B11.4, Shears, Construction, Care, and Use, and ANSI B11.3., Power Press Brakes, Construction, Care, and use(DoD adopted).

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Provisions suitable for hoisting the ironworker shall be provided. The base shall provide a means for securely bolting the ironworker to the floor. The ironworker shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2 Controls. The press shall be equipped with all the controls required for ironworker operations. All operating controls shall be located and designed for use by a single operator. With the exception of the Type I ironworker which shall be operated by a three-position footswitch, the Type II and Type III ironworkers shall be controlled independently by either a five-position joystick control or three-position footswitch. The joystick control must be capable of returning the ironworker to a neutral position when released from any position stopping the ironworker from any further movement. The footswitch shall be used by inserting wired plug end into a receptacle located on the front and rear control boxes.

3.3 Drive Assembly. The ironworker shall be provided with a mechanical and hydraulic drive. The drive shall include a drive motor, hydraulic pump, hydraulic filter , hydraulic cylinder, and a crank arm assembly. The drive assembly shall be mounted inside the ironworker and accessible by an access door. All hydraulic components shall be approved by the National Fluid Power Association (NFPA).

3.3.1 Drive motor. The manufacturers standard motor provided to the commercial market shall be supplied. The electrical system shall conform to the ANSI/National Fire Protection Association (ANSI/NFPA) 79 (see Table II).

3.3.2 Hydraulic Pump. The hydraulic pump shall consist of a sealed system with a cylinder being able to achieve the requirements in Table II. The hydraulic pump shall be mounted to the electrical drive motor and must have a changeable filter assembly. A hydraulic pressure gauge shall be included with the Type II and III equipment.

3.3.3 Hydraulic Filter. A minimum of a 4" Pall style filter assembly with a changeable element shall be provided. One additional filter element shall be included.

3.3.4 Hydraulic Cylinder. The hydraulic cylinder shall be capable of achieving ton requirements in Table II. The assembly shall have all fittings and attachments required for immediate operation and be capable of rebuild by a seal kit. Two seal kits shall be included.

3.3.5 Crank Arm. The crank arm shall be of adequate size and material to prevent deflection when subjected to maximum torque. The crank arm shall be mounted on hinge pins and bushings of a size, type, and precision class suitable for the application.

TABLE II. Minimum ironworker parameters and performance requirements.

Parameters / Performance		Type I	Type II	Type III
Hydraulic System	Drive Motor (hp)	3	10	10
	Drive Motor, Electrical System	220/440v	220/440v	220/440v
	Hydraulic Tank Capacity	12 gallons	25 Gallons	25 Gallons
Working Surface Area	Throat Depth	5"	10"	12"
	Punch End Table	8" x 16"	12" x 20"	14.5" x 21
	Coping End Table	12" x 14"	18" x 18"	19" x 19"

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TABLE II. Minimum ironworker parameters and performance requirements. Continued

Machine Capacities (60 ksi tensile mild steel)	Punching (Max Hole Ø / Max Material)	13/16" Ø / ¾" Thick 3" Ø / 3/16" Thick Maximum 50 Tons	1-1/8" Ø / 1" Thick 5" Ø / 3/16" Thick Maximum 90 Tons	1-7/16" Ø / 1" Thick 5" Ø / 1/4" Thick Maximum 110 Tons
	Bending	Maximum 50 Tons	Maximum 90 Tons	Maximum 110 Tons
	Bar Shear (Min – Max)	¼" – 1 3/8" Round Stock ½" - 1" Square Stock	½' - 2" Round Stock ½" – 1 3/8" Square Stock	½" - 2" Round Stock ½" - 1 3/8" Square Stock
	Plate Shear (Max width x thickness)	13" x ¼" 12" x 3/8" 10 x ½" 6" x ¾"	18" x ½" 12" x ¾" 8" x 1"	20" x ½" 18" x ¾" 12" x 1"
	Angle Shear	3" x 3" x 3/8"	5" x 5" x ½"	6" x 6" x 5/8"
	Coper-Notcher	2 ¼" x 3" x ¼"	4" x 6" x 3/8"	4" x 6" x 3/8"

3.4 Standard equipment. All standard equipment normally provided with the manufacturers commercial press shall be furnished. Furthermore, any other accessories identified in this document that is not part of the standard equipment shall also be provided.

3.5 Punch station. The punch station shall consist of a punch holder, die holder, and stripper. The punch station shall be capable of punching holes of the sizes specified (see Table II). The station shall have a throat depth suitable to accommodate the material size specified see 7.1(a). Means shall be provided to lower the punch for positioning over the workpiece.

3.5.1 Punch holder. The ironworker shall be provided with a punch holder for holding the punch in position under the ram. The punch holders shall be constructed of alloy steel and/or cast steel and securely hold the punch in position. The punch holders shall assure alignment of the punches with the dies. The punch holders shall be designed to accommodate standard round and shaped hole punches.

3.5.2 Die holder. The ironworker shall be provided with a die holder for holding the die in position. The die holder shall be constructed of hardened steel and/or cast steel precision ground to securely hold the dies in such a manner to prevent the dies from rotating. The die holder shall be designed to accommodate standard round and shaped hole dies.

3.5.3 Stripper. The ironworker shall be provided with a stripper for automatically releasing the workpiece from the punch when the punch is withdrawn from the die. The stripper shall permit changing the punch without removing the stripper from the ironworker.

3.5.4 Punches and dies. At a minimum, the following punch and die size shall be provided with the equipment, 9/16", 11/16", 13/16, 15/16", and 1-1/16". The punches and dies shall be tool steel, hardened, and ground for accuracy. Each punch and die shall be permanently marked for size and identification.

3.6 Bending attachment. Bending attachments shall be included in the following sizes: 6" for Type I, 12" for Type II, and 14" for Type III.

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3.6 Notcher station. The notcher station shall be capable of notching rectangular or triangular cuts in metal to meet the requirements specified (See Table II). The notcher shall be equipped with an adjustable gauge designed to govern the depth of the cut. A guard for the notcher shall provide for protecting the operator when other stations aside from the notcher are being used. The bladed of the notcher shall be replaceable.

3.7 Flat bar shear station. The bar shear station shall consist of an upper moveable blade and a lower stationary blade and shall be capable of shearing the maximum width flat bar stock specified in in Table II. The shear blades shall be mounted in such a manner that they can be replaced without the use of special tools. The flat bar shear shall be equipped with a stripper or hold down which shall be adjustable to accommodate stock of any dimension within the capacity of the ironworker. The hold down shall be no more than 4 inches from the shearing plane.

3.8 Feed table. A material support and feed table shall be provided. The feed table shall be equipped with rollers to facilitate the movement of material and shall be a minimum of 5 feet long. The table shall be capable of being mounted directly to the ironworker.

3.9 Power requirements. Unless otherwise specified, each press shall be designed to operate on 3-phase, 60-Hertz (Hz) 220/440-volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components and swapping of some overload and fuse components. The ironworker shall be initially wired as specified in Table II.

3.10 Motors. Motors shall be the manufacturer's standard motor supplied to the commercial market and shall meet horsepower and electrical design, and operational requirements as specified in Table II.

3.11 Size and Capacity. The ironworker shall meet the size and capacity requirements as specified in Table I.

3.12 Physical size limitations. When specified (see 7.1b), the physical size of the press shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the press will fit into its future operation location.

3.13 Marking on instruments, control panels, charts, and plates. All words on the press, including all instruments, control panels, charts, plates, and related attachments, shall be in the English language. Characters shall be marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.14 Data plates. All standard data plates normally provided with the manufacturers standard commercial press shall be furnished (see 7.1c).

3.15 Nameplate. A nameplate shall be securely attached to each press. At minimum, the nameplate shall contain the information listed below.

a. Nomenclature

b. Manufacturer's name

c. Machine model designation

d. Serial number

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4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

6. PACKAGING.

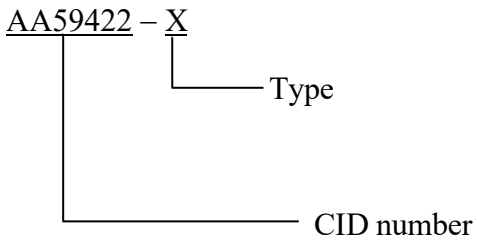
6.1 Preservation, packing, and marking. For acquisition purposes, the ironworker shall be preserved, packed, and marked as specified in the acquisition order (see 7.1e).

7. NOTES.

7.1 Ordering data. Acquisition documents must specify the following information:

a. CID document number, revision, and CID PIN
b. Physical size limitations, if required (see 3.12)
c. Additional data plate, if required (see 3.14)
d. Nameplate requirements (see 3.15)
e. Packaging requirements (see 6.1)

7.2 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



7.3 Sources of documents.

7.3.1 Government documents. Copies of specifications, standards, and handbooks are available from DLS Document Services, online at https://assist.dla.mil.

7.3.2 American national Standards Institute (ANSI) documents. Copies of ANSI documents may be obtained at https://ansi.org.

7.3.3 National Fire Protection Association (NFPA). NFPA documents may be obtained at https://www.nfpa.org/.

7.3.4 Federal Acquisition regulation (FAR). Copies of FAR may be obtained at https://www.acquisition.gov/browse/index/far.

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